# Rossmann Sales Forecasting

**Presented by**

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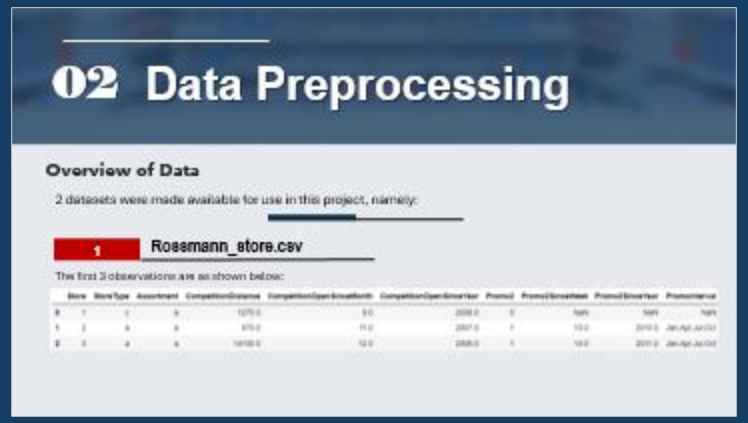


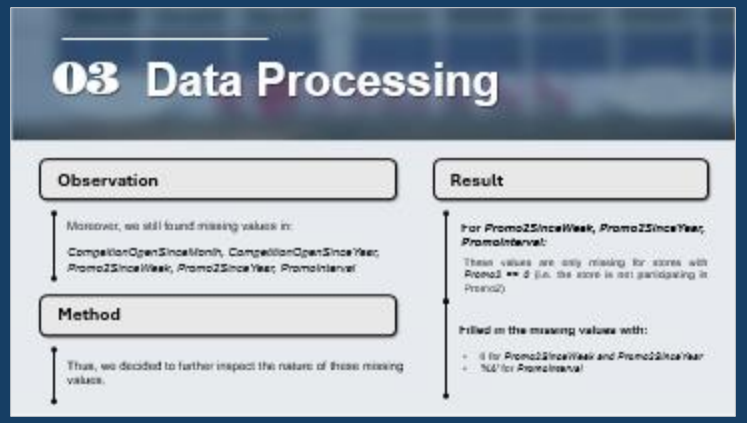


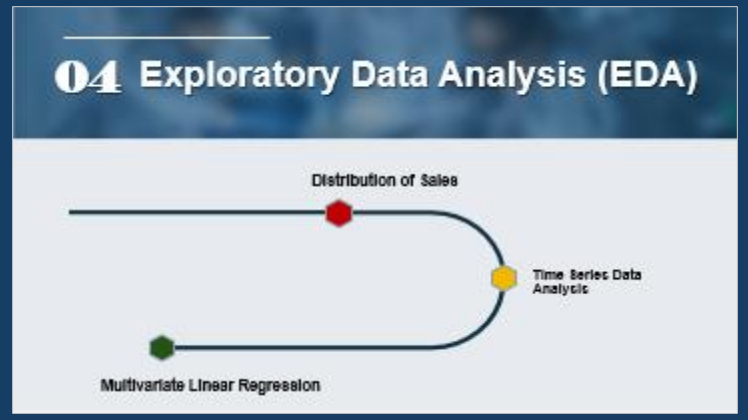

# 01 Introduction

## Background of Rossmann drug store:

### Foundation

Rossmann is one of the largest drug store chains in Europe. Based in Germany, it was founded in the year of 1972.

### Product

The company offers a diverse selection of personal care, household items, and pharmaceuticals.

### Presence

With over 4,000 stores, Rossmann focuses on providing affordable, quality products to customers primarily in European countries.

# 01 Introduction

## Their desire

## Understand and forecast trends

As a consumer-facing business, sales are a major indicator of customer satisfaction and effective business management. By understanding and forecasting sales trends, store managers can better plan resources and meet customer demands.

# 01 Introduction

## Research questions

1

### What would the trend of sales be like?

We aim to develop a machine learning model capable of predicting daily sales for Rossmann Stores.

2

### What factors affect sales?

We aim to develop a machine learning model capable of estimating the number of sales based on the potential factors.

# 02 Data Preprocessing

## Overview of Data

2 datasets were made available for use in this project, namely:

### 1 Rossmann_store.csv

The first 3 observations are as shown below:

| | Store | StoreType | Assortment | CompetitionDistance | CompetitionOpenSinceMonth | CompetitionOpenSinceYear | Promo2 | Promo2SinceWeek | Promo2SinceYear | PromoInterval |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | c | a | 1270.0 | 9.0 | 2008.0 | 0 | NaN | NaN | NaN |
| **1** | 2 | a | a | 570.0 | 11.0 | 2007.0 | 1 | 13.0 | 2010.0 | Jan,Apr,Jul,Oct |
| **2** | 3 | a | a | 14130.0 | 12.0 | 2006.0 | 1 | 14.0 | 2011.0 | Jan,Apr,Jul,Oct |

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### It contains the details of:

**1115** stores AND **10** variables*

* 10 variables : *Store, Store type, Assortment, CompetitionDistance, CompetitionOpenSinceMonth, CompetitionOpenSinceYear, Promo2, Promo2SinceWeek, Promo2SinceYear, PromoInterval*

* Missing values found in: *CompetitionDistance, CompetitionOpenSinceMonth, CompetitionOpenSinceYear, Promo2SinceWeek, Promo2SinceYear, PromoInterval*

# Overview of Data

2 datasets were made available for use in this project, namely:

## 2 Rossmann_train.csv

The first 3 observations are as shown below:

| | Store | DayOfWeek | Date | Sales | Customers | Open | Promo | StateHoliday | SchoolHoliday |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 5 | 2015-07-31 | 5263 | 555 | 1 | 1 | 0 | 1 |
| **1** | 2 | 5 | 2015-07-31 | 6064 | 625 | 1 | 1 | 0 | 1 |
| **2** | 3 | 5 | 2015-07-31 | 8314 | 821 | 1 | 1 | 0 | 1 |

## It contains the historical sales data:

**1,017,209** observations AND **9** variables*

* 9 variables : *Store, DayOfWeek, Date, Sales, Customers, Open, Promo, StateHoliday, SchoolHoliday*

* No missing values exist in any of the columns.

# Our Approach

For this project, we focused on analysing the top 20% of stores in terms of sales.

## Filtering & Merging

### Calculate Average Sales

We calculated the average sales for each store over the 31-month period in the data *(+ add this information into the store dataframe)*

### Obtain top 20% of stores

Next, we obtained the store IDs of the top 20% of stores in terms of sales from the store dataframe *(avg. sales ≥ 80% quantile)*

### Filtering

Filtered both the store dataset & sales dataset by the Store IDs obtained

### Merging

Finally, we merged the store dataframe & sales dataframe on the Store IDs

## Result

Therefore, we got..

### 1 A Single Dataframe

A single dataframe, containing the past sales data of 223 stores

### 2 Complete Variables

*18 variables* with *Date* set as index

# 02 Data Preprocessing

## Overview of Preprocessed Data

The first 3 observations of the combined data frame:

| Date | Store | StoreType | Assortment | CompetitionDistance | CompetitionOpenSinceMonth | CompetitionOpenSinceYear | Promo2 | Promo2SinceWeek | Promo2SinceYear | Promo2Interval | DayOfWeek | Sales | Customers | Open | Promo | StateHoliday | SchoolHoliday |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2015-07-31 | 4 | c | c | 620.0 | 9.0 | 2009.0 | 0 | 0 | 0 | NA | 5 | 13995.0 | 1498 | 1 | 1 | 0 | 1 |
| 2015-07-31 | 7 | a | c | 24000.0 | 4.0 | 2013.0 | 0 | 0 | 0 | NA | 5 | 15344.0 | 1414 | 1 | 1 | 0 | 1 |
| 2015-07-31 | 24 | a | c | 4590.0 | 3.0 | 2000.0 | 1 | 40 | 2011 | Jan,Apr,Jul,Oct | 5 | 14190.0 | 1082 | 1 | 1 | 0 | 1 |

# 03 Data Processing

## Observation

Moreover, we still found missing values in:

*CompeitionOpenSinceMonth, CompetitionOpenSinceYear, Promo2SinceWeek, Promo2SinceYear, PromoInterval*

## Method

Thus, we decided to further inspect the nature of these missing values.

## Result

**For *Promo2SinceWeek, Promo2SinceYear, PromoInterval:***

These values are only missing for stores with *Promo2 == 0* (i.e. the store is not participating in Promo2)

**Filled in the missing values with:**

- 0 for *Promo2SinceWeek and Promo2SinceYear*
- *'NA'* for *PromoInterval*

# 03 Data Processing

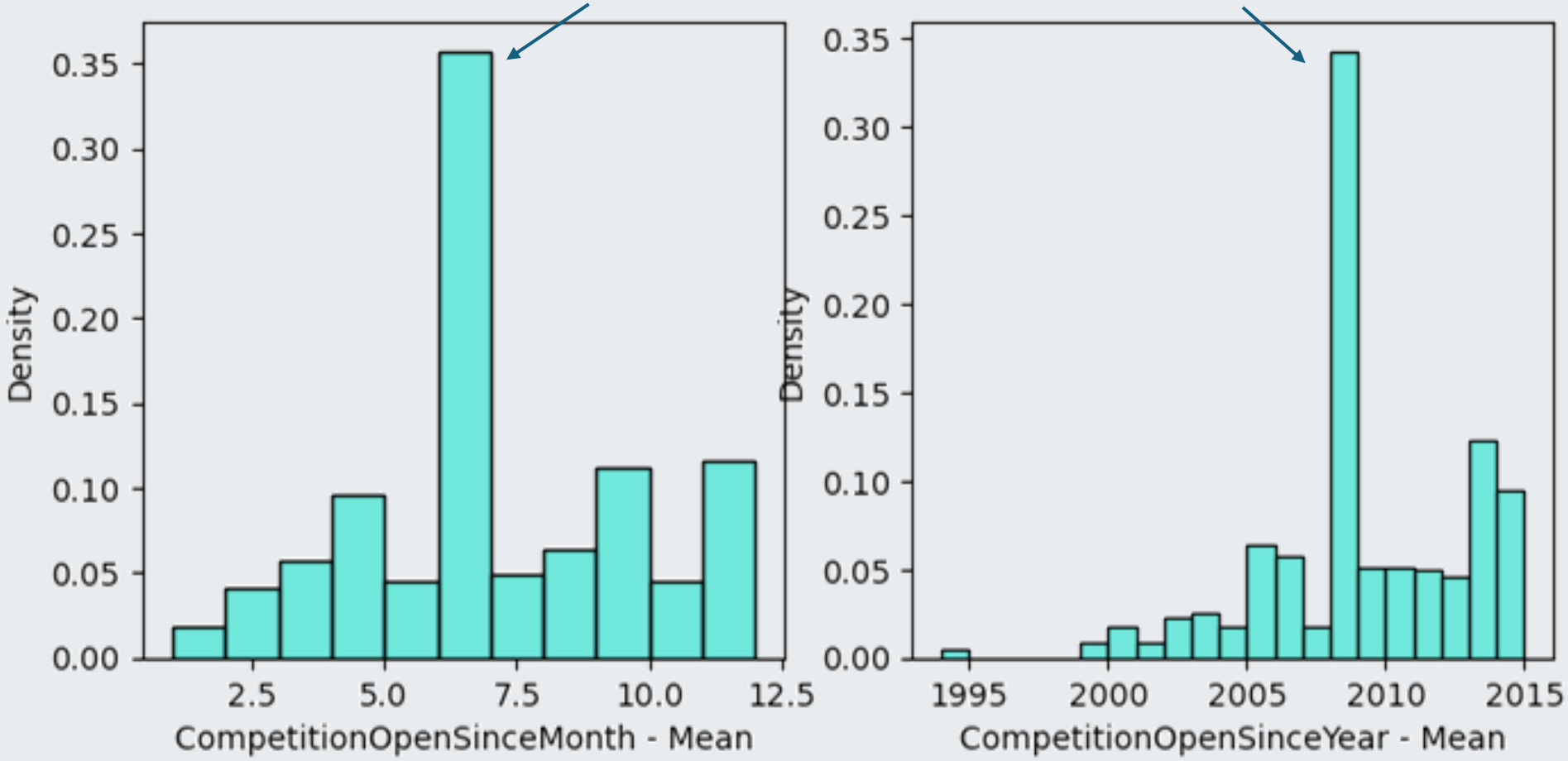
## Imputation

1. Simple Imputation
2. kNN Imputation

# 1 Simple Imputation

Initially, we filled in missing values using **mean** and **median**:

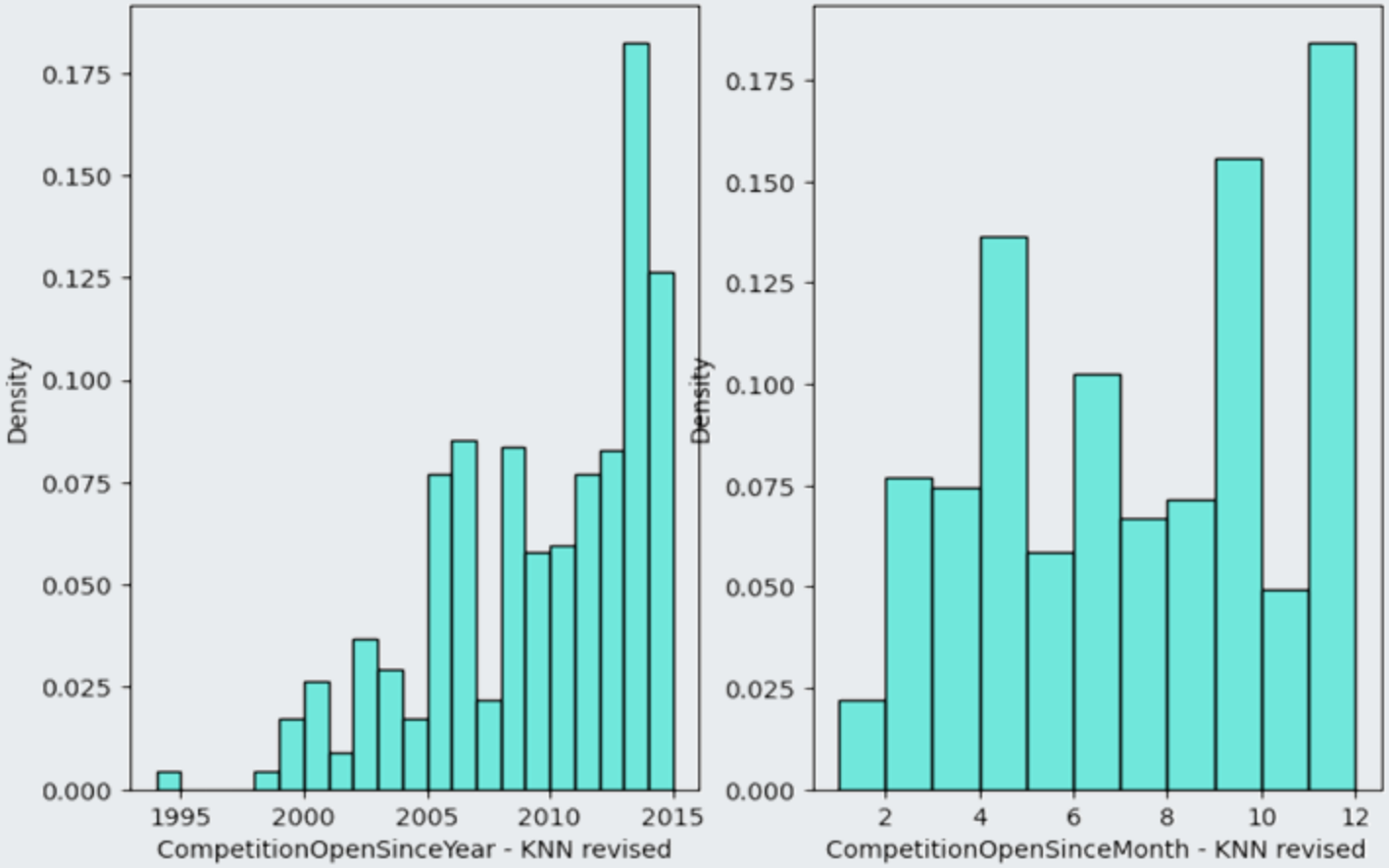
Obvious peak shown in histogram…



# 2 kNN Imputation

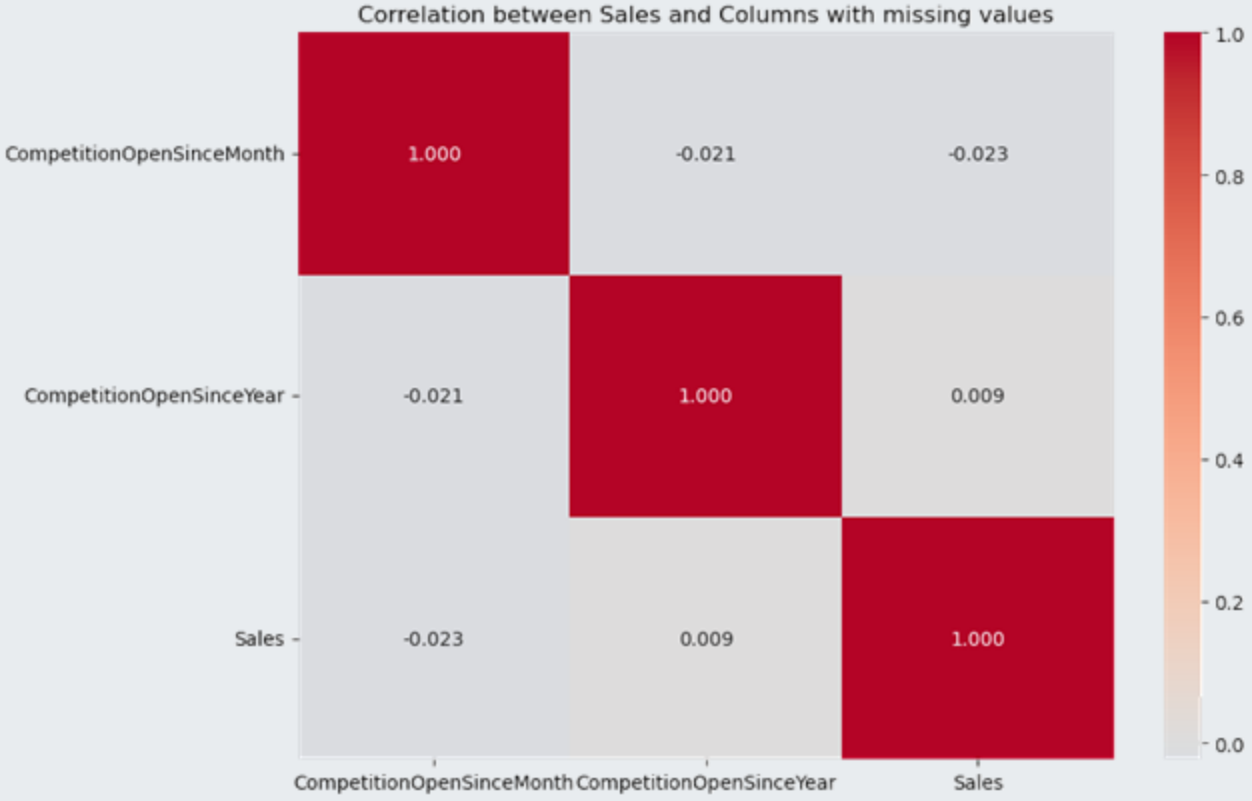
Then, we tried out using kNN imputation:

A **bell shape** curve histogram!

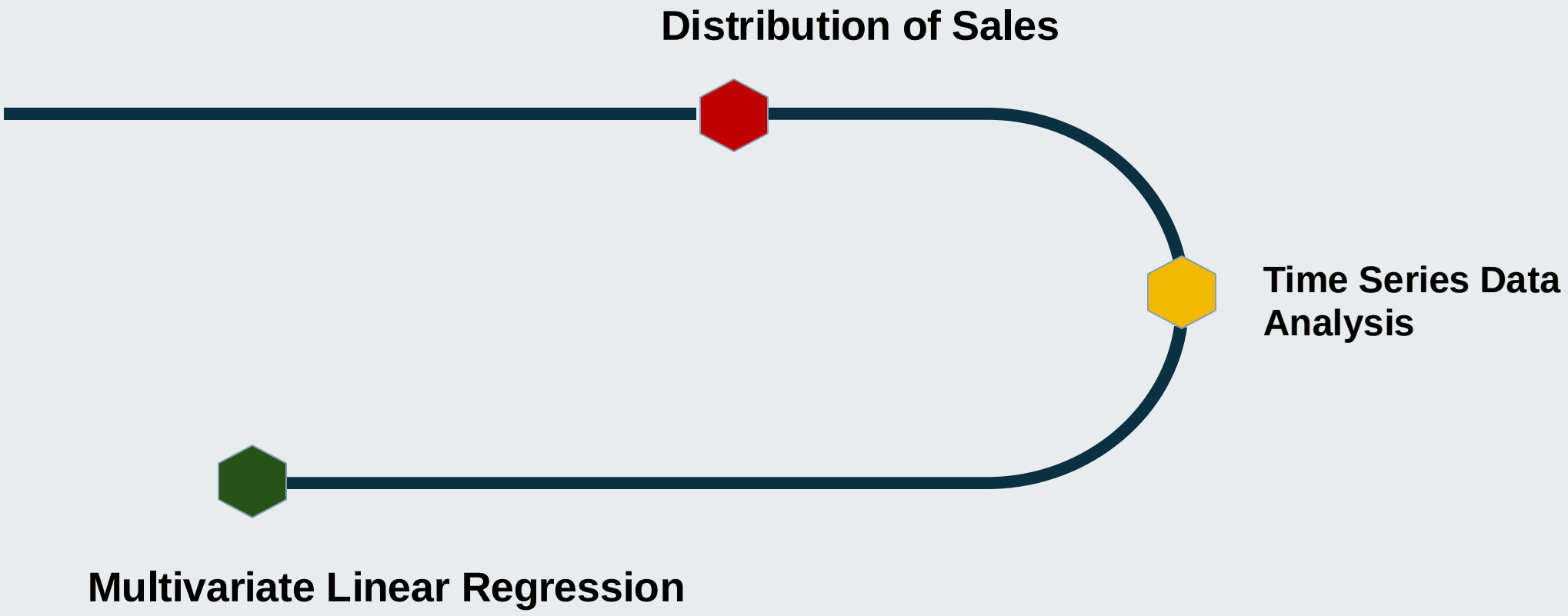


# 2 kNN Imputation

The bell shape curve is not perfect as **CompetitionOpenSinceYear** and **CompetitionOpenSinceMonth** has very weak correlation with target variable '**Sales**'.

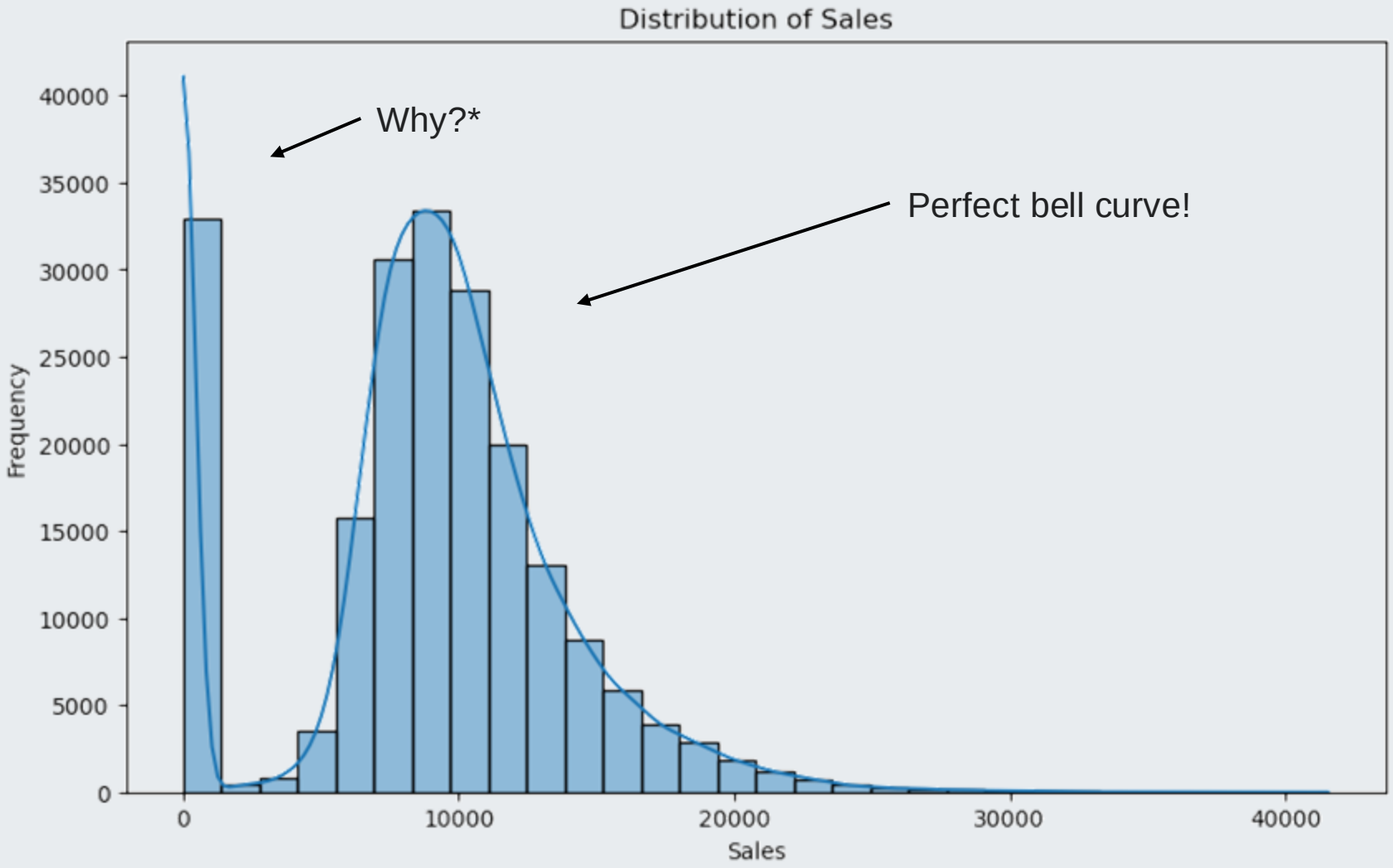


# 04 Exploratory Data Analysis (EDA)



# Distribution of Sales

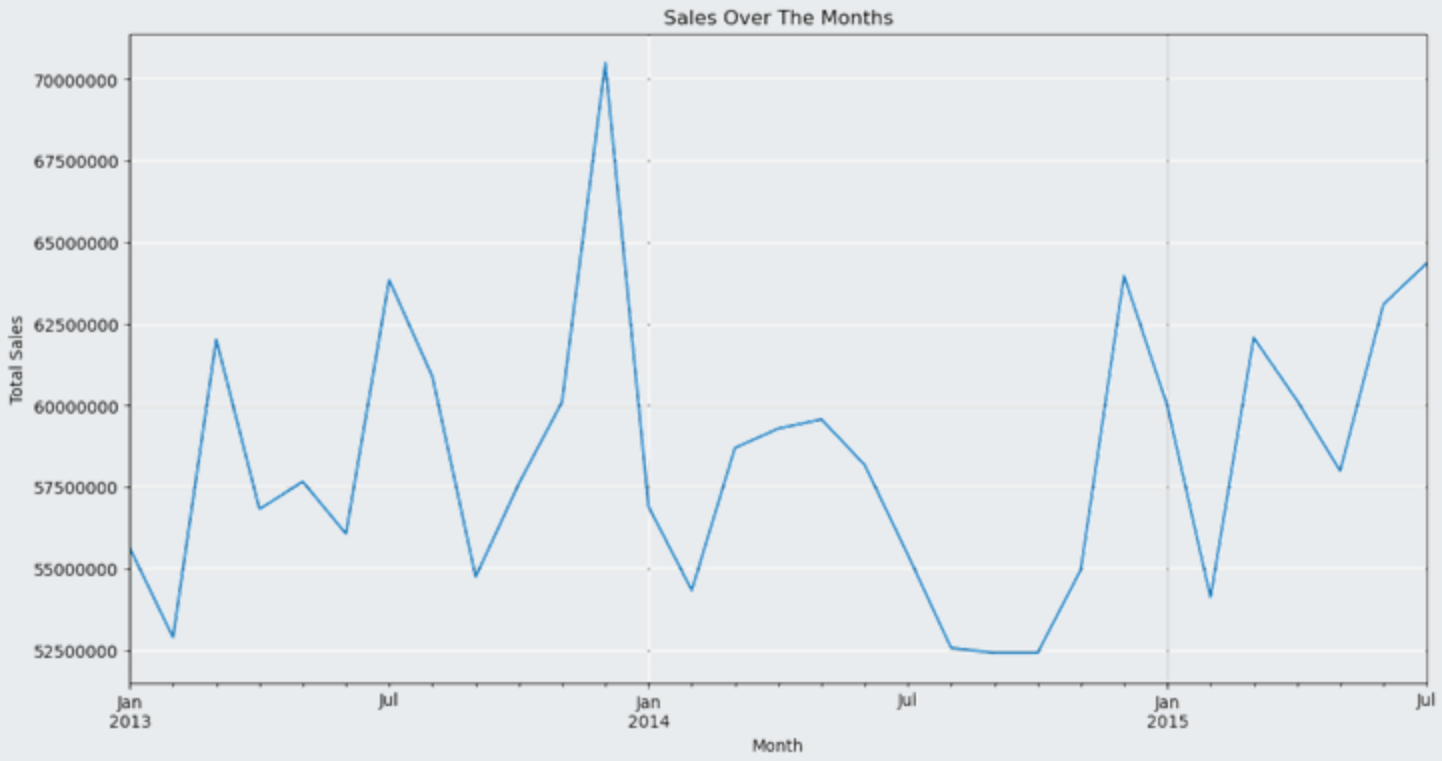
We visualised the distribution of sales using a histogram.




*This is common in retail datasets where certain stores may not have sales on specific days or during certain periods (not open).

# Time Series Data Analysis – Over Months

We visualised the trend of sales for each month.

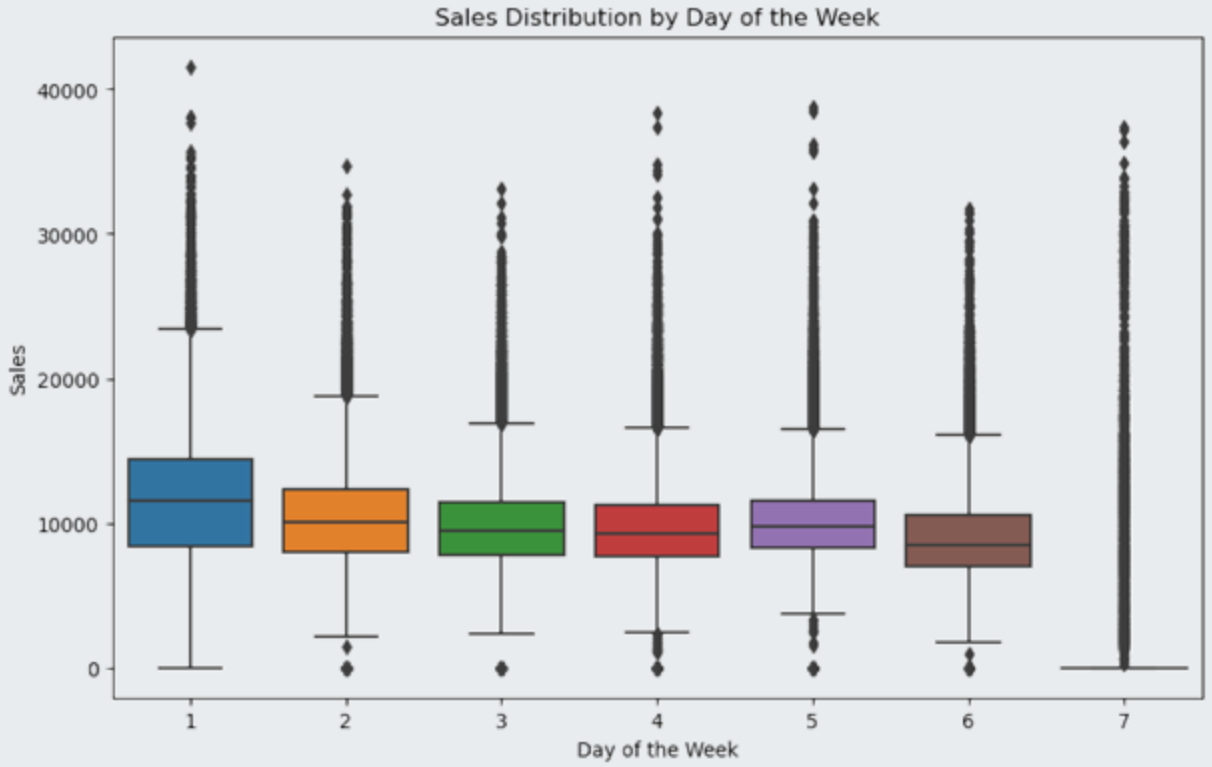



**Observation:**

Sales peaked in December 2013, declined until October 2014, then rose again in December 2014.

# Time Series Data Analysis – Over Week

We visualised the distribution of sales for each day of the week.




**Observations :**

- Monday has the highest median sales and wider variability.
- Sunday shows the lowest median and extreme outliers.
- Tuesday to Friday exhibit consistent sales with a few outliers, particularly on Monday and Saturday.

# Time Series Data Analysis – Rolling Mean

Next, we used the rolling mean technique to investigate the effect of Promo.

**Observations :**

- Aligning with our expectations, promotions seemed to have a noticeably positive effect on sales.
- This is because lower prices would make products more affordable and motivate consumers to get the most of their money's worth.

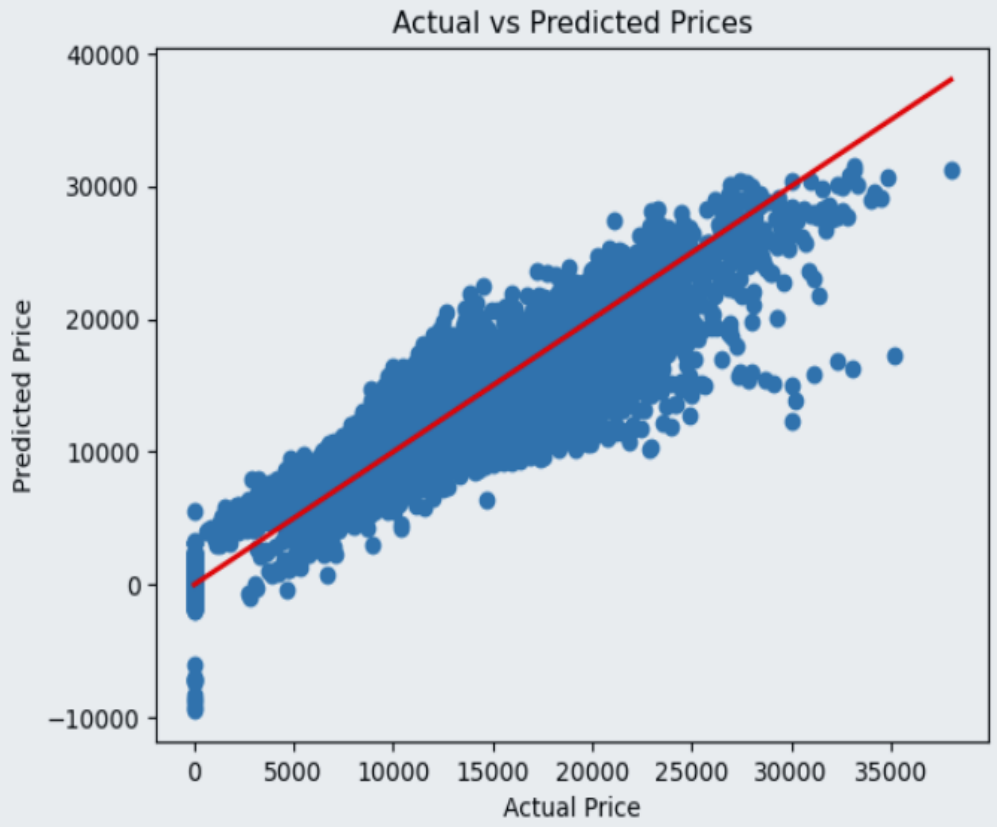
# Multivariate Linear Regression

We have numerous of factors that may affect sales.

***Which of them are the key factors?***

# Multivariate Linear Regression

We did some simple modelling to test the feasibility of the MLR model




```
Intercept: 11569.80
Mean Squared Error: 2340614.57
R-squared: 0.91
```

This model could not be accepted as,

## It receives noise.

0.91 of R-squared value indicates that it overfits. It is receiving all variables regardless of relevance. Thus, it does not generalise well to the testing data.

## Large intercept values

The intercept suggested that if all factors influencing sales were at their baseline levels, we would still set the baseline sales value as 11569.80. This is unreasonable.

# Multivariate Linear Regression

Thus, we did a few approaches to simplify our model. These includes:

1. Analysis of Correlation
2. Analysis of Coefficients
3. Regularisation

# Multivariate Linear Regression

## 1 Analysis of Correlation

### What we did

We filtered out variables with correlations less than 0.2.

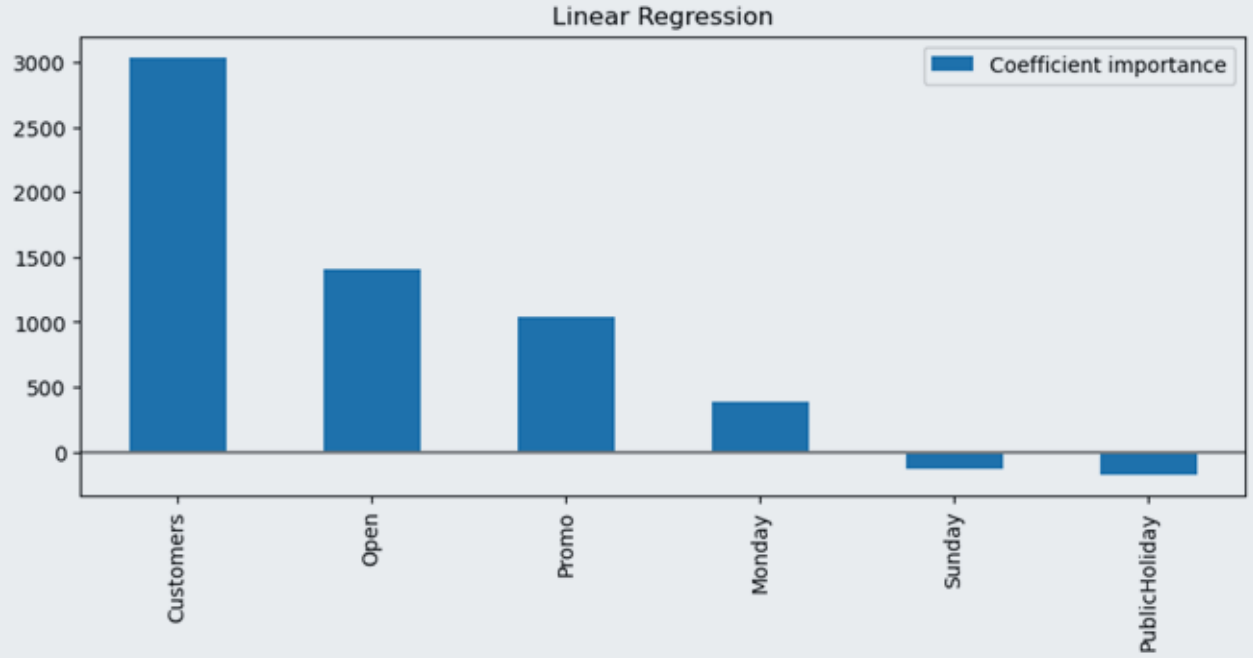
### Result

We left with variables below:

- *PublicHoliday*
- *Sunday*
- *Monday*
- *Promo*
- *Open*
- *Customers*

# Multivariate Linear Regression

## 2 Analysis of Coefficients




### What we did

We investigated the importance of model coefficients by its coefficient values and standard deviation.

### Result

This suggested that *Customers*, *Open* and *Promo* are the top 3 most important variables contributing to sales.

# Multivariate Linear Regression

## 3 Regularisation

### What we did

**First,**

We normalised all the data using:

$$nX = X - X.mean\,(\,)/\ X.std(\,)$$

We determined whether we should be using Lasso or Ridge regularisation methods.

We considered that,

- Feature selection is important
- Interpretability is Crucial
- Strong Feature Correlation

Thus, we chose to use:

**Lasso Regularisation**

# Multivariate Linear Regression

## Lasso Regularisation

### Hyperparameter tuning

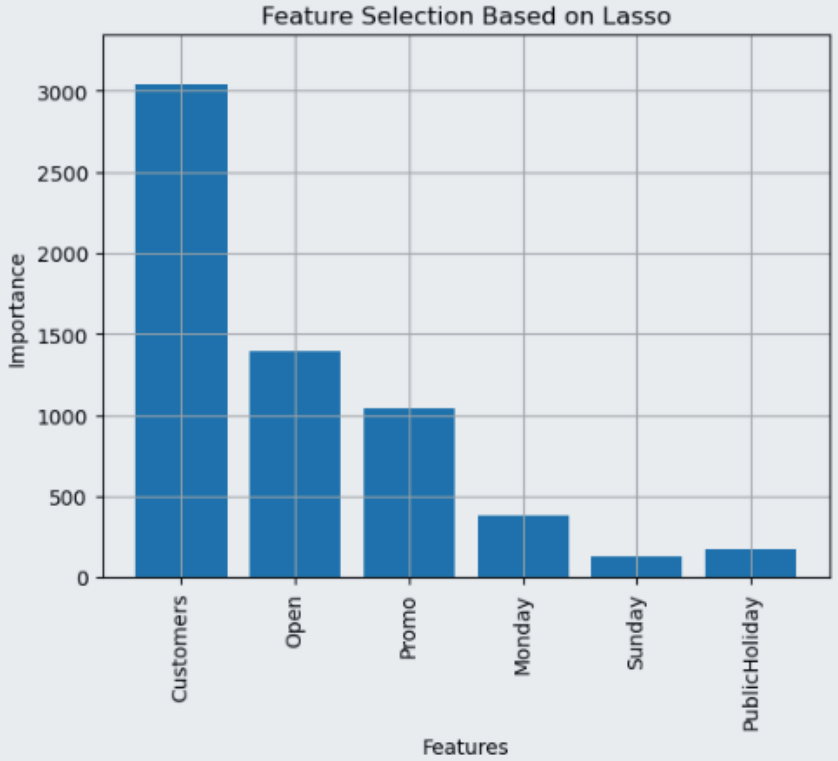
To find the optimal value of regularisation parameter `alpha`.

We generated 50 values between:

$$10^{-4} \quad \text{AND} \quad 10^{1}$$

to find the optimal value of regularisation parameter `alpha`. In this case,

**Optimal alpha value is:** 0.0176

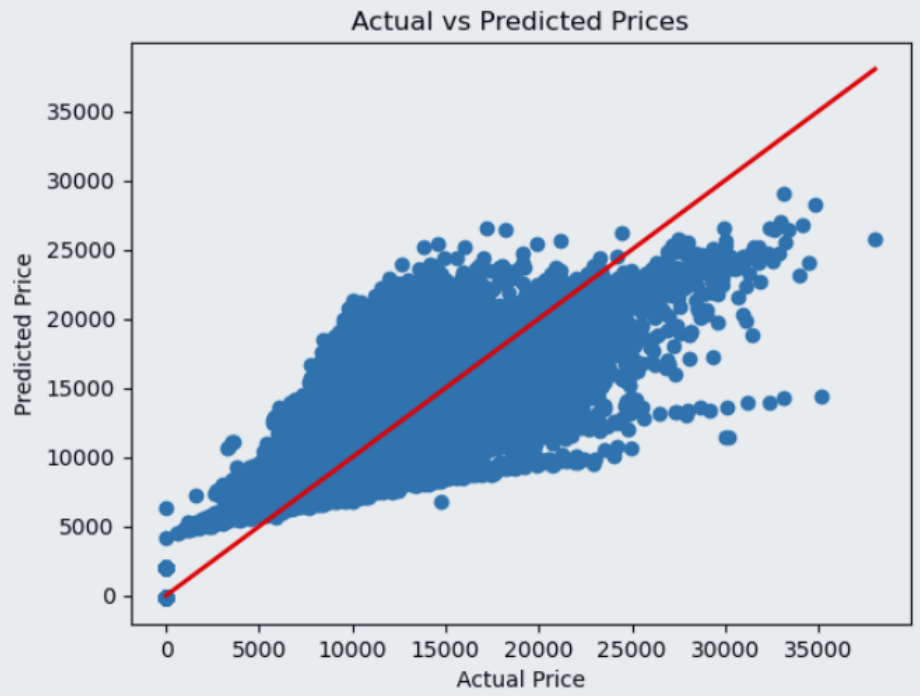



## Result

This suggested that *Customers*, *Open* and *Promo* are the top 3 most important variables contributing to sales.

# Multivariate Linear Regression

Our final result is as below:




```
Coefficients: ['3062.65', '1581.79', '1065.77']
Intercept: 8807.13
Mean Squared Error: 3886632.94
R-squared: 0.85
```

Comparing to the initial model,

## Its generalisability increased.

85% of the variance in the sales data is explained by the model.

## Reasonable initial baseline

The value of baseline decreased, which is near to the value it supposed to be.

## Final Result:

$$Sales = 8807.13 + 3062.85x_1 + 1581.79x_2 + 1065.77x_3$$

$Sales_{pred}$ stands for predicted sales, $x_1$ stands for number of customers, $x_2$ stands for the availability of the store, and $x_3$ stands for the availability of promotion.

# 05 Predictive Modelling

Recall of our main aim:

***What are the predictions for the overall drugstore sales four weeks in advance?***

Thus, we have:

## Training set

All the past sales data except the final 4 weeks

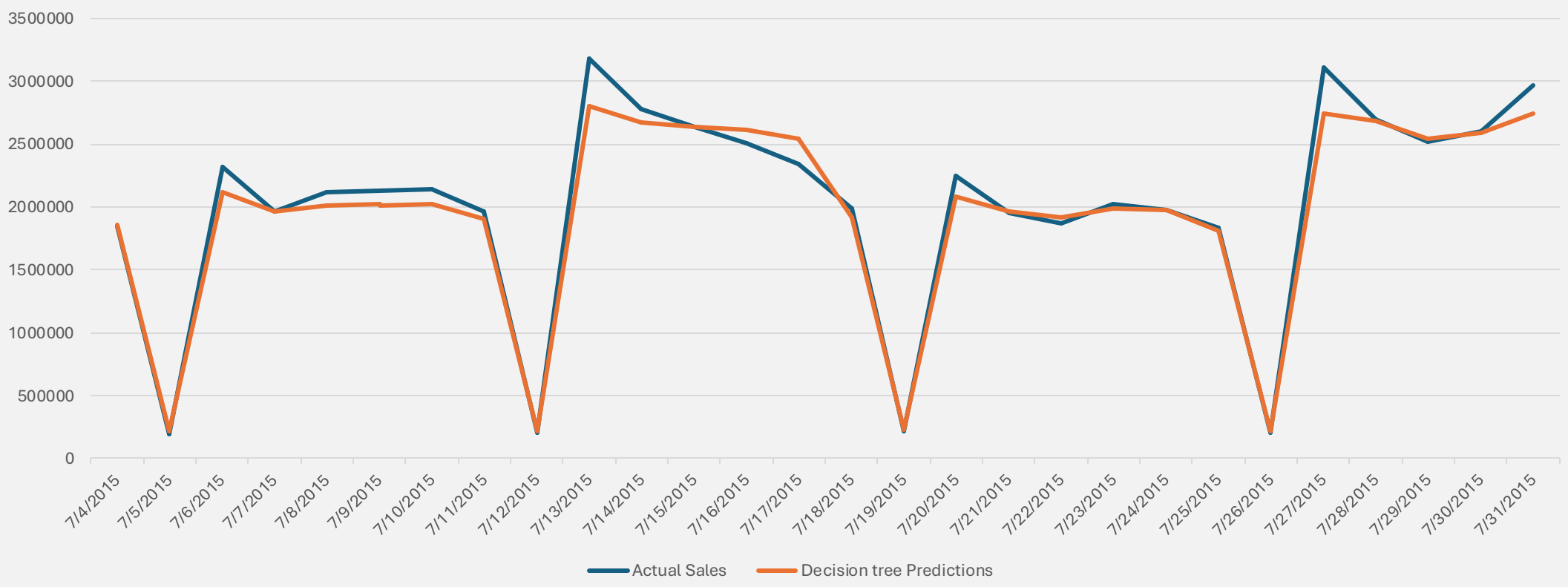
## Testing set

The sales data of the last 4 weeks in the dataset

# Single Regression Tree

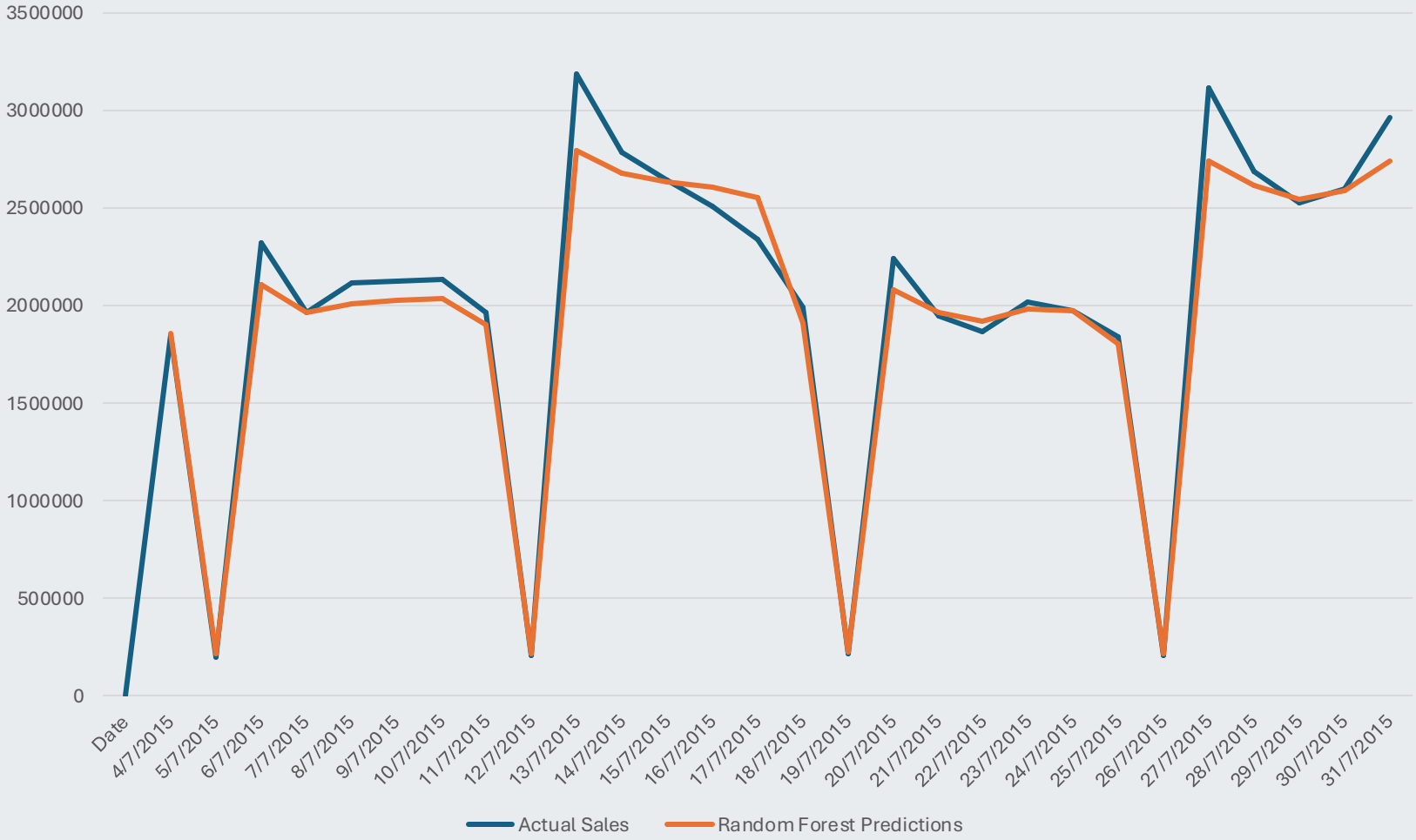
*Optimal* **max_depth** *giving least difference between training and testing score:* **7**

```
Training score: 0.86, Testing score: 0.84
```

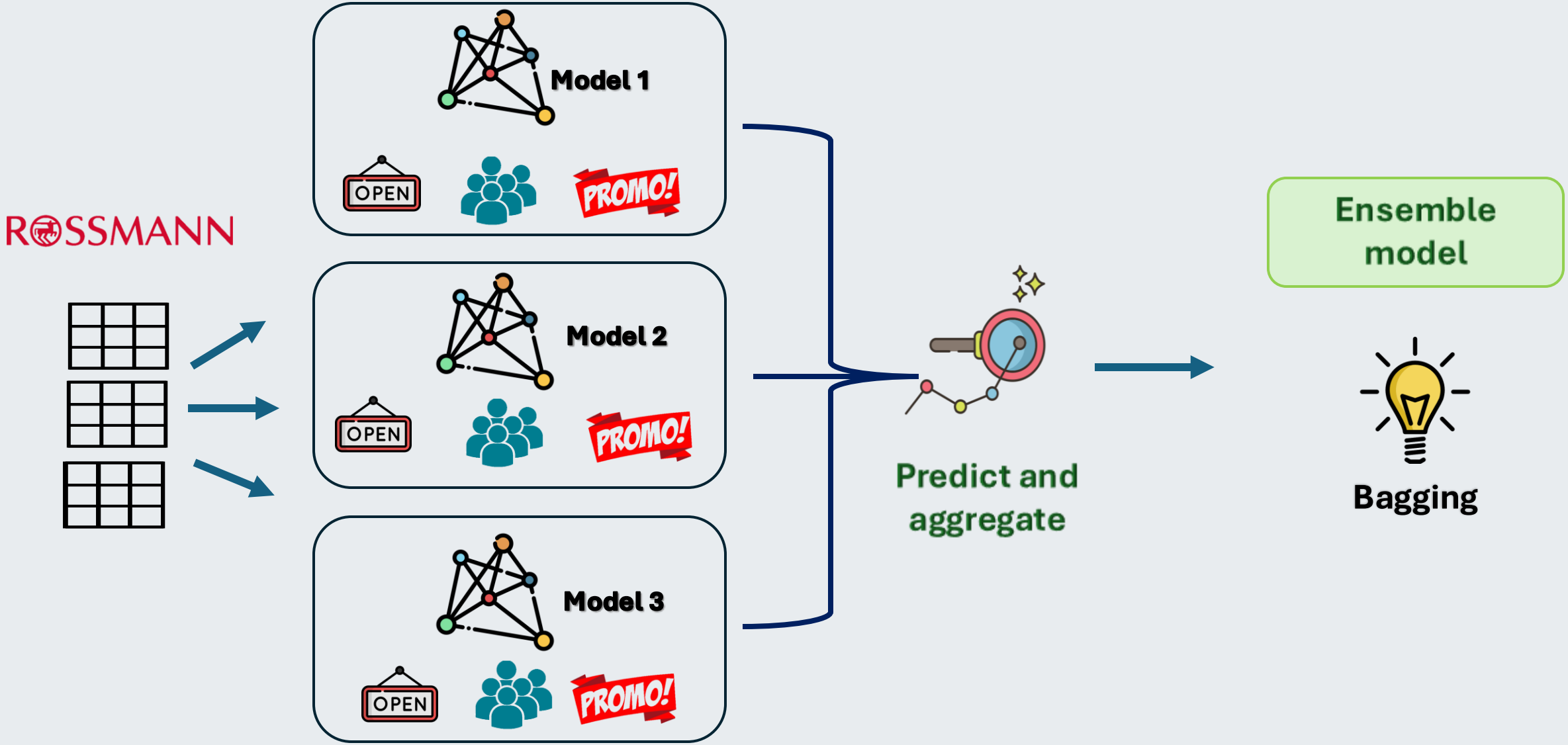


# Ensemble learning- Random Forest Regression

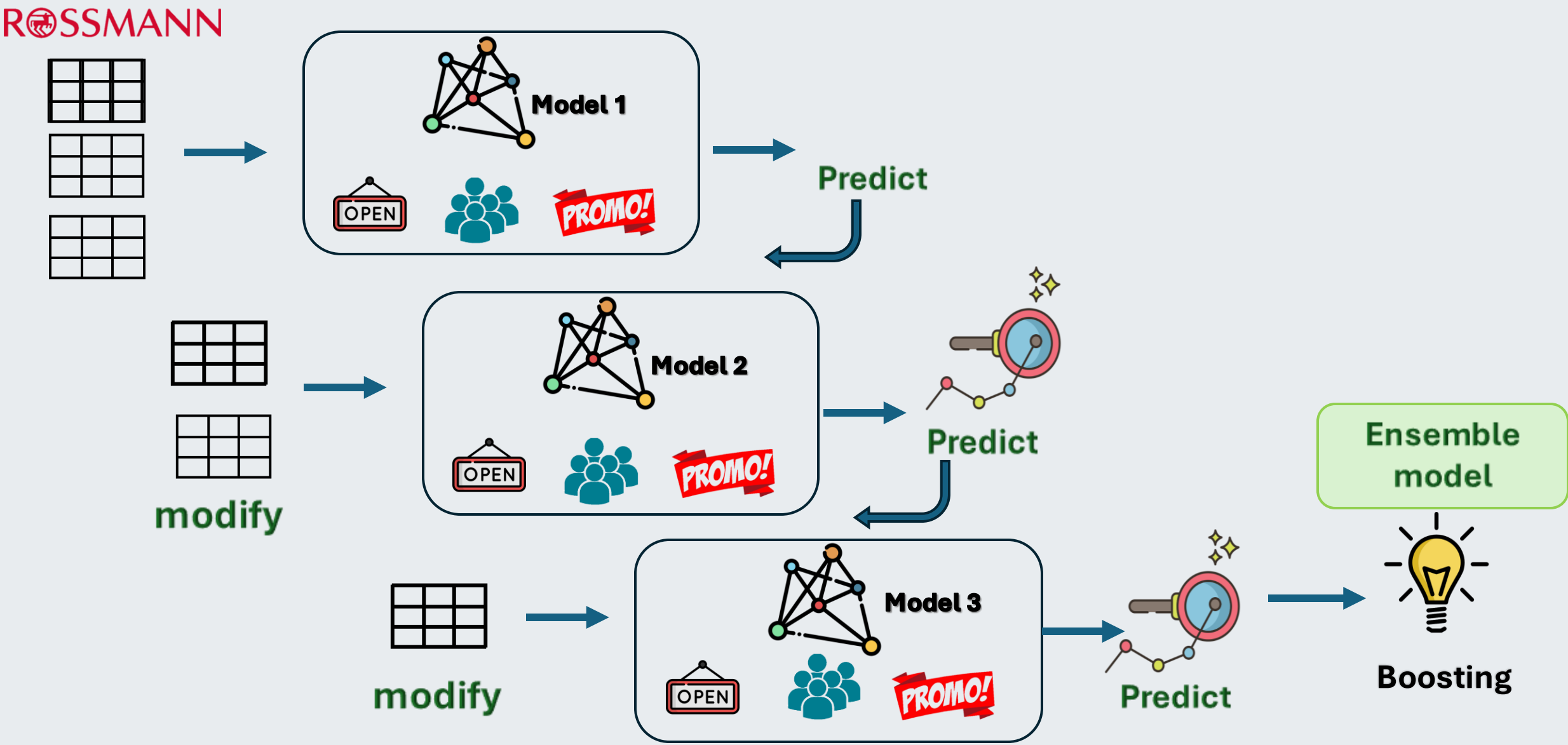
| | R^2 | RMSE | MAE |
|---|---|---|---|
| train | 0.862026 | 1899.599467 | 1230.237359 |
| test | 0.838858 | 1927.604685 | 1246.977878 |



# Ensemble learning – Bagging & Boosting



# Ensemble learning – Boosting



# Ensemble learning – Boosting

## AdaBoostRegressor

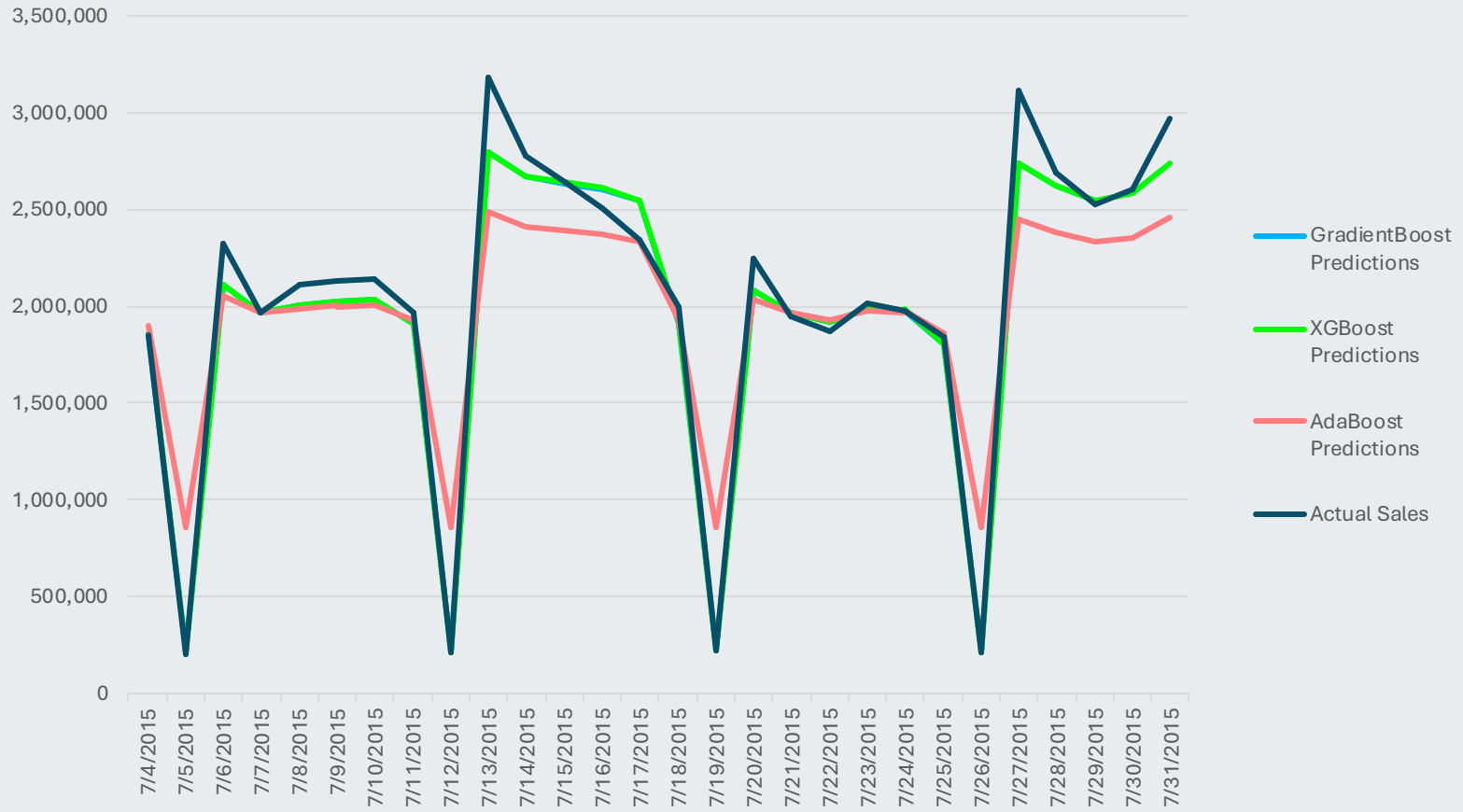
| | R^2 | RMSE | MAE |
|---|---|---|---|
| train | 0.861033 | 1906.423430 | 1238.787319 |
| test | 0.840104 | 1920.135348 | 1246.718493 |

## GradientBoostRegressor

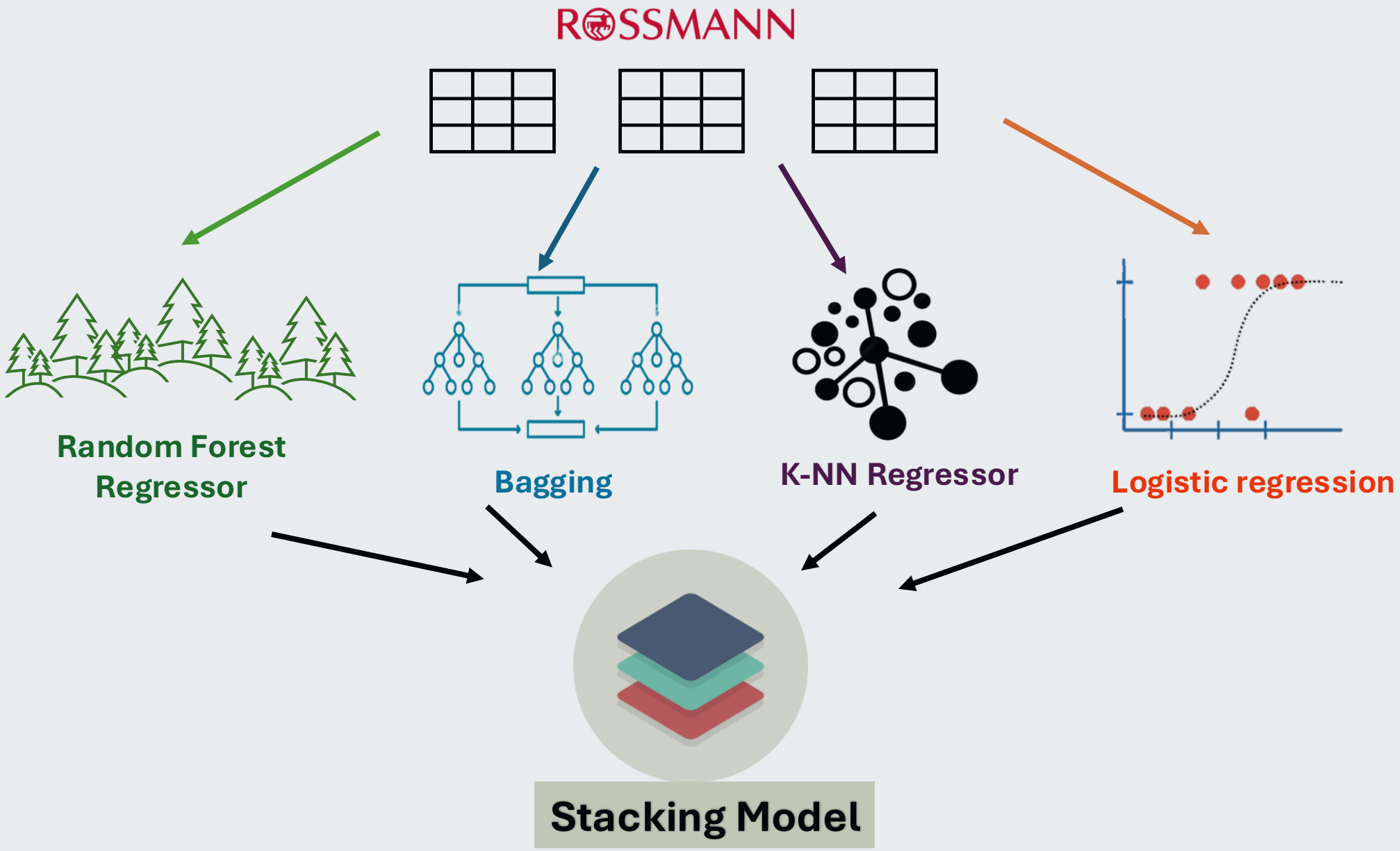
| | R^2 | RMSE | MAE |
|---|---|---|---|
| train | 0.860339 | 1911.178702 | 1244.316960 |
| test | 0.839802 | 1921.948979 | 1251.552007 |

## XGBoost

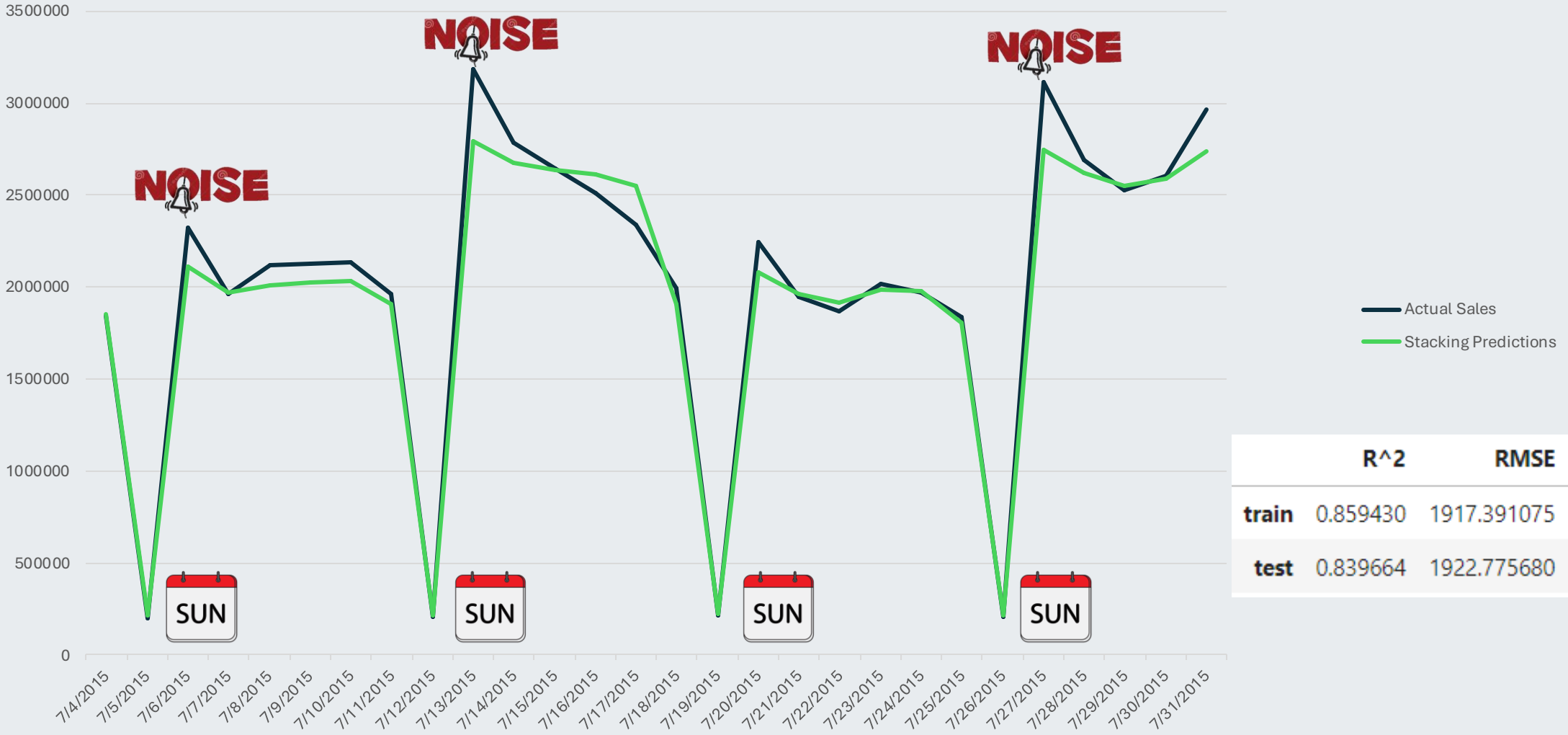
| | R^2 | RMSE | MAE |
|---|---|---|---|
| train | 0.857752 | 1928.797622 | 1243.596359 |
| test | 0.837598 | 1935.123993 | 1249.317690 |



# Ensemble learning – Stacking



# Ensemble learning – Stacking




| | R^2 | RMSE | MAE |
|---|---|---|---|
| **train** | 0.859430 | 1917.391075 | 1239.852966 |
| **test** | 0.839664 | 1922.775680 | 1245.911777 |

# 06 Conclusion

## Our Findings

### Exploratory Data Analysis (EDA)

Sales peaked in **Dec 2013**

***Monday*** has the highest median sales

### Features Selection

***Key factors:***

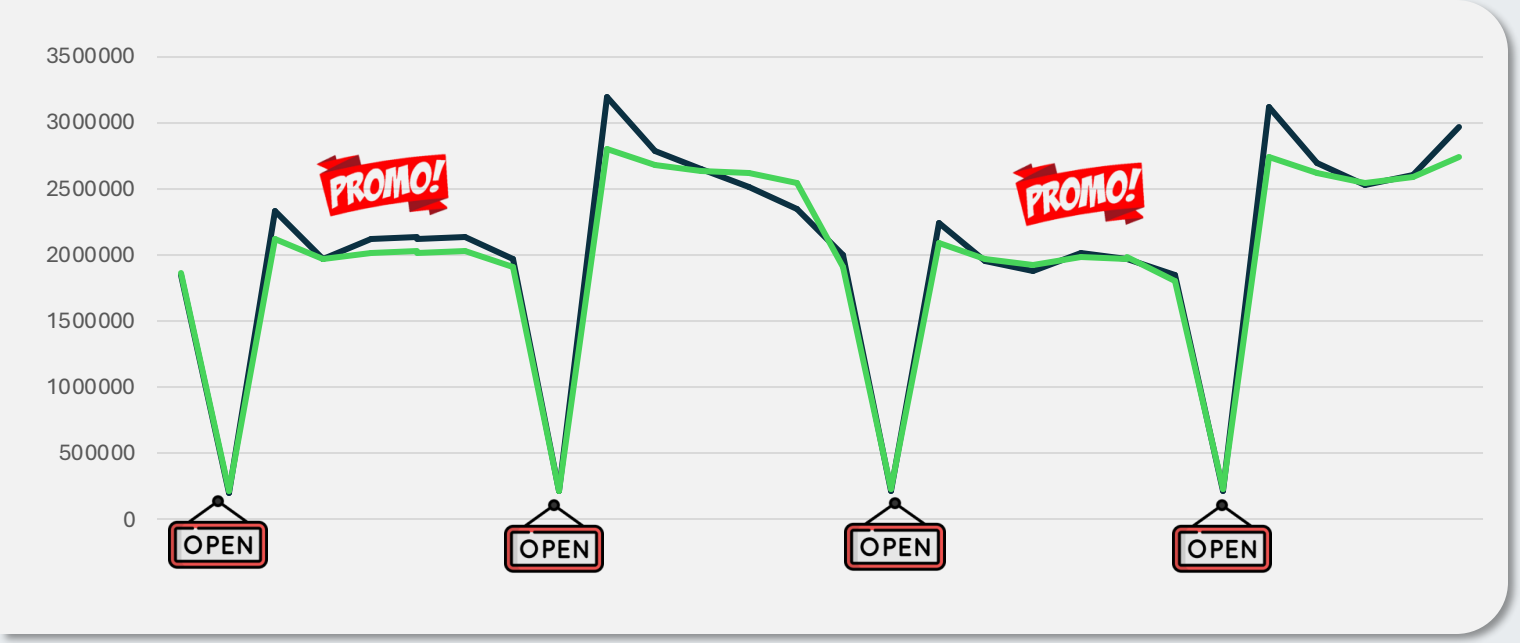
***Open*** ***Promo*** ***Customer***

### Predictive modeling

**Best model:**

**Stacking model**

# Our Suggestions

1. Increase the frequency for store **opening**, ideally everyday.

2. Increase the frequency of **promo events**.